

# KIT SAFETY DATA SHEET



**Kit Product Name** ReadyPrep Protein Extraction Kit (Total)  
**Kit Catalog Number(s)** 1632086  
**Revision date** 15-Mar-2023

## Kit Contents

| Catalog Number(s)            | Product Name                              |
|------------------------------|-------------------------------------------|
| 1632101, 1632101EDU, 9703632 | ReadyPrep TBP Reducing Agent              |
| 1632083, 10009795            | ReadyPrep 2-D Rehydration/Sample Buffer 1 |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 23-Aug-2022

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name ReadyPrep TBP Reducing Agent

### Other means of identification

Catalog Number(s) 1632101, 1632101EDU, 9703632

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### **Corporate Headquarters**

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### **Manufacturer Address**

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### **Legal Entity / Contact Address**

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                                                  |             |
|--------------------------------------------------|-------------|
| Skin corrosion/irritation                        | Category 2  |
| Serious eye damage/eye irritation                | Category 2A |
| Reproductive toxicity                            | Category 1B |
| Specific target organ toxicity (single exposure) | Category 3  |
| Flammable liquids                                | Category 4  |

### Label elements

#### **Danger**

#### **Hazard statements**

Causes skin irritation  
Causes serious eye irritation  
May damage fertility or the unborn child  
May cause respiratory irritation  
Combustible liquid

**Precautionary Statements - Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

Wash face, hands and any exposed skin thoroughly after handling

Avoid breathing dust/fume/gas/mist/vapors/spray

Use only outdoors or in a well-ventilated area

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

**Precautionary Statements - Response**

IF exposed or concerned: Get medical advice/attention

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

**Skin**

IF ON SKIN: Wash with plenty of water and soap

If skin irritation occurs: Get medical advice/attention

Take off contaminated clothing and wash it before reuse

**Inhalation**

IF INHALED: Remove person to fresh air and keep comfortable for breathing

Call a POISON CENTER or doctor if you feel unwell

**Fire**

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Precautionary Statements - Storage**

Store in a well-ventilated place. Keep container tightly closed

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

May be harmful if swallowed. May be harmful if inhaled.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name          | CAS No   | Weight-% | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|------------------------|----------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1-Methyl-2-pyrrolidone | 872-50-4 | 50 - 100 | -                                                                                     |                                                                   |

## 4. First-aid measures

### Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General advice                     | Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                            |
| Inhalation                         | Remove to fresh air. IF exposed or concerned: Get medical advice/attention. Get medical attention immediately if symptoms occur.                                                                                                                                                    |
| Eye contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists. |
| Skin contact                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if irritation develops and persists.                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Call a physician.                                                                                                                       |
| Self-protection of the first aider | Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Wear personal protective clothing (see section 8). Avoid contact with skin, eyes or clothing.      |

### Most important symptoms and effects, both acute and delayed

|          |                                                               |
|----------|---------------------------------------------------------------|
| Symptoms | May cause redness and tearing of the eyes. Burning sensation. |
|----------|---------------------------------------------------------------|

### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

## 5. Fire-fighting measures

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Dry chemical. Carbon dioxide (CO <sub>2</sub> ). Water spray. Alcohol resistant foam.                                              |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray.        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | Yes.                                                                                                                               |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                      |                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions | Evacuate personnel to safe areas. Use personal protective equipment as required. See section 8 for more information. Take precautionary measures against static discharges. Do not touch or walk through spilled material. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. |
| Other information    | Refer to protective measures listed in Sections 7 and 8.                                                                                                                                                                                                                                           |

### Methods and material for containment and cleaning up

|                                |                                                                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods for containment</b> | Stop leak if you can do it without risk. Do not touch or walk through spilled material. Dike far ahead of liquid spill for later disposal.                 |
| <b>Methods for cleaning up</b> | Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers. |

## 7. Handling and storage

### Precautions for safe handling

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Use personal protection equipment. Do not breathe vapor or mist. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Take precautionary measures against static discharges. Use with local exhaust ventilation. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Remove contaminated clothing and shoes. Take off contaminated clothing and wash before reuse. In case of insufficient ventilation, wear suitable respiratory equipment. |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labeled containers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Store locked up. Store according to product and label instructions. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name                      | Alberta | British Columbia | Ontario                    | Quebec |
|------------------------------------|---------|------------------|----------------------------|--------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 |         |                  | TWA: 400 mg/m <sup>3</sup> |        |

### Appropriate engineering controls

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

### Individual protection measures, such as personal protective equipment

|                                       |                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>            | Tight sealing safety goggles.                                                                                                                                                                                                                                                                                                                                  |
| <b>Hand protection</b>                | Wear suitable gloves. Impervious gloves.                                                                                                                                                                                                                                                                                                                       |
| <b>Skin and body protection</b>       | Wear suitable protective clothing. Long sleeved clothing.                                                                                                                                                                                                                                                                                                      |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.                                                                                                                                                                                       |
| <b>General hygiene considerations</b> | Do not eat, drink or smoke when using this product. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Wear suitable gloves and eye/face protection. Avoid contact with skin, eyes or clothing. |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | Liquid                   |
| Color          | colorless                |
| Odor           | Amine                    |
| Odor threshold | No information available |

| <u>Property</u>                        | <u>Values</u>       | <u>Remarks • Method</u> |
|----------------------------------------|---------------------|-------------------------|
| pH                                     |                     | None known              |
| Melting point / freezing point         | -24 °C / -11.2 °F   |                         |
| Boiling point / boiling range          | No data available   | None known              |
| Flash point                            | 90 °C / 194 °F      |                         |
| Evaporation rate                       | No data available   | None known              |
| Flammability (solid, gas)              | No data available   | None known              |
| Flammability Limit in Air              |                     | None known              |
| Upper flammability or explosive limits | No data available   |                         |
| Lower flammability or explosive limits | No data available   |                         |
| Vapor pressure                         | No data available   | None known              |
| Vapor density                          | No data available   | None known              |
| Relative density                       | No data available   | None known              |
| Water solubility                       | Immiscible in water |                         |
| Solubility in other solvents           | No data available   | None known              |
| Partition coefficient                  | No data available   | None known              |
| Autoignition temperature               | 270 °C / 518 °F     | None known              |
| Decomposition temperature              |                     | None known              |
| Kinematic viscosity                    | No data available   | None known              |
| Dynamic viscosity                      | No data available   | None known              |

### Other information

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

## 10. Stability and reactivity

|                                    |                                                      |
|------------------------------------|------------------------------------------------------|
| Reactivity                         | No information available.                            |
| Chemical stability                 | Stable under normal conditions.                      |
| Possibility of hazardous reactions | None under normal processing.                        |
| Conditions to avoid                | Heat, flames and sparks.                             |
| Incompatible materials             | Strong acids. Strong bases. Strong oxidizing agents. |
| Hazardous decomposition products   | None known based on information supplied.            |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|            |                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. May be harmful if inhaled. |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|

|                     |                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Irritating to eyes. (based on components). Causes serious eye irritation.                                    |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes skin irritation. (based on components).                                                               |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. May be harmful if swallowed. |

#### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Redness. May cause redness and tearing of the eyes.

#### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

|                                      |                 |
|--------------------------------------|-----------------|
| <b>ATEmix (oral)</b>                 | 3,324.90 mg/kg  |
| <b>ATEmix (dermal)</b>               | 26,190.50 mg/kg |
| <b>ATEmix (inhalation-dust/mist)</b> | 5.32 mg/l       |

#### Component Information

| Chemical name                      | Oral LD50            | Dermal LD50         | Inhalation LC50        |
|------------------------------------|----------------------|---------------------|------------------------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 | = 3914 mg/kg ( Rat ) | = 8 g/kg ( Rabbit ) | > 5.1 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                                                                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Irritating to skin.                                                                         |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye irritation.                                                              |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met.                                                                                   |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met.                                                                                   |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met.                                                                                   |
| <b>Reproductive toxicity</b>             | Contains a known or suspected reproductive toxin. Classification based on data available for ingredients. May damage fertility or the unborn child. |
| <b>STOT - single exposure</b>            | May cause respiratory irritation.                                                                                                                   |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met.                                                                                   |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met.                                                                                   |

## 12. Ecological information

#### Ecotoxicity

| Chemical name                      | Algae/aquatic plants                             | Fish                                                                                                                                           | Toxicity to microorganisms | Crustacea                               |
|------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 | EC50: >500mg/L (72h,<br>Desmodesmus subspicatus) | LC50: =832mg/L (96h,<br>Lepomis macrochirus)<br>LC50: =1072mg/L (96h,<br>Pimephales promelas)<br>LC50: =1400mg/L (96h,<br>Poecilia reticulata) | -                          | EC50: =4897mg/L (48h,<br>Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Component Information**

| Chemical name                      | Partition coefficient |
|------------------------------------|-----------------------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 | -0.46                 |

**Other adverse effects** No information available.

### 13. Disposal considerations

**Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

### 15. Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

### 16. Other information

|                                   |                           |                                  |                           |                                           |
|-----------------------------------|---------------------------|----------------------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 2   | <b>Flammability</b> 2            | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b>                       | <b>Health hazards</b> 2 * | <b>Flammability</b> 2            | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend</i> |                           | <i>* = Chronic Health Hazard</i> |                           |                                           |

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend** Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION



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|                |                                                    |           |                                                      |
|----------------|----------------------------------------------------|-----------|------------------------------------------------------|
| TWA<br>Ceiling | TWA (time-weighted average)<br>Maximum limit value | STEL<br>* | STEL (Short Term Exposure Limit)<br>Skin designation |
|----------------|----------------------------------------------------|-----------|------------------------------------------------------|

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Aug-2022

**Revision Note** Reformatted and updated existing information.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 23-Aug-2022

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name ReadyPrep 2-D Rehydration/Sample Buffer 1

### Other means of identification

Catalog Number(s) 1632083, 10009795

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### **Corporate Headquarters**

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### **Manufacturer Address**

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### **Legal Entity / Contact Address**

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                       |             |
|-----------------------|-------------|
| Carcinogenicity       | Category 1B |
| Reproductive toxicity | Category 2  |

### Label elements

#### **Danger**

#### **Hazard statements**

May cause cancer  
Suspected of damaging fertility or the unborn child

**Precautionary Statements - Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

IF exposed or concerned: Get medical advice/attention

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

May be harmful if swallowed. May be harmful in contact with skin. Toxic to aquatic life with long lasting effects. Toxic to aquatic life.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | IF exposed or concerned: Get medical advice/attention. Show this safety data sheet to the doctor in attendance.      |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                 |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| <b>Skin contact</b>   | Wash skin with soap and water.                                                                                       |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

## 5. Fire-fighting measures

|                                                       |                                                                                                                                    |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                 | No information available.                                                                                                          |
| <b>Specific hazards arising from the chemical</b>     | None known.                                                                                                                        |
| <b>Explosion data</b>                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                | None.                                                                                                                              |
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                             |                                                          |
|-----------------------------|----------------------------------------------------------|
| <b>Personal precautions</b> | See section 8 for more information.                      |
| <b>Other information</b>    | Refer to protective measures listed in Sections 7 and 8. |

### Methods and material for containment and cleaning up

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| <b>Methods for containment</b> | Prevent further leakage or spillage if safe to do so. |
| <b>Methods for cleaning up</b> | Pick up and transfer to properly labeled containers.  |

## 7. Handling and storage

### Precautions for safe handling

|                                |                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Remove contaminated clothing and shoes. |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                           |                                                                     |
|---------------------------|---------------------------------------------------------------------|
| <b>Storage Conditions</b> | Store locked up. Store according to product and label instructions. |
|---------------------------|---------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

|                        |                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b> | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Appropriate engineering controls

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                |                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection            | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| Hand protection                | Wear suitable gloves.                                                                                                                                                    |
| Skin and body protection       | Wear suitable protective clothing.                                                                                                                                       |
| Respiratory protection         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| General hygiene considerations | Do not eat, drink or smoke when using this product. Wash hands before breaks and immediately after handling the product.                                                 |

**9. Physical and chemical properties****Information on basic physical and chemical properties**

|                |                          |
|----------------|--------------------------|
| Physical state | Solid                    |
| Appearance     | solid                    |
| Color          | white                    |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| <u>Property</u>                        | <u>Values</u>     | <u>Remarks • Method</u> |
|----------------------------------------|-------------------|-------------------------|
| pH                                     | 10                |                         |
| Melting point / freezing point         | No data available | None known              |
| Boiling point / boiling range          | No data available | None known              |
| Flash point                            | No data available | None known              |
| Evaporation rate                       | No data available | None known              |
| Flammability (solid, gas)              | No data available | None known              |
| Flammability Limit in Air              |                   | None known              |
| Upper flammability or explosive limits | No data available |                         |
| Lower flammability or explosive limits | No data available |                         |
| Vapor pressure                         | No data available | None known              |
| Vapor density                          | No data available | None known              |
| Relative density                       | No data available | None known              |
| Water solubility                       | Soluble in water  |                         |
| Solubility in other solvents           | No data available | None known              |
| Partition coefficient                  | No data available | None known              |
| Autoignition temperature               | No data available | None known              |
| Decomposition temperature              |                   | None known              |
| Kinematic viscosity                    | No data available | None known              |
| Dynamic viscosity                      | No data available | None known              |

**Other information**

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

**10. Stability and reactivity**

|                                    |                                 |
|------------------------------------|---------------------------------|
| Reactivity                         | No information available.       |
| Chemical stability                 | Stable under normal conditions. |
| Possibility of hazardous reactions | None under normal processing.   |

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| Conditions to avoid              | None known based on information supplied. |
| Incompatible materials           | None known based on information supplied. |
| Hazardous decomposition products | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | May be harmful in contact with skin.                              |
| Ingestion    | May be harmful if swallowed.                                      |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

|                 |                |
|-----------------|----------------|
| ATEmix (oral)   | 4,300.30 mg/kg |
| ATEmix (dermal) | 2,889.50 mg/kg |

#### Component Information

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met.                                                                                              |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met.                                                                                              |
| Respiratory or skin sensitization | Based on available data, the classification criteria are not met.                                                                                              |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met.                                                                                              |
| Carcinogenicity                   | Contains a known or suspected carcinogen. Classification based on data available for ingredients. May cause cancer.                                            |
| Reproductive toxicity             | Contains a known or suspected reproductive toxin. Classification based on data available for ingredients. Suspected of damaging fertility or the unborn child. |
| STOT - single exposure            | Based on available data, the classification criteria are not met.                                                                                              |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met.                                                                                              |
| Aspiration hazard                 | Based on available data, the classification criteria are not met.                                                                                              |

## 12. Ecological information

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Ecotoxicity | Toxic to aquatic life. Toxic to aquatic life with long lasting effects. |
|-------------|-------------------------------------------------------------------------|

|                               |                                    |
|-------------------------------|------------------------------------|
| Persistence and degradability | No information available.          |
| Bioaccumulation               | There is no data for this product. |
| Other adverse effects         | No information available.          |

### 13. Disposal considerations

#### Waste treatment methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

### 14. Transport information

|                        |               |
|------------------------|---------------|
| TDG                    | Not regulated |
| DOT                    | Not regulated |
| UN number or ID number | UN3077        |
| Packing group          | III           |

### 15. Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

### 16. Other information

|                                   |                                  |                |                    |                                    |
|-----------------------------------|----------------------------------|----------------|--------------------|------------------------------------|
| <b>NFPA</b>                       | Health hazards 0                 | Flammability 0 | Instability 0      | Physical and chemical properties - |
| <b>HMIS</b>                       | Health hazards 1*                | Flammability 0 | Physical hazards 0 | Personal protection X              |
| <i>Chronic Hazard Star Legend</i> | <i>* = Chronic Health Hazard</i> |                |                    |                                    |

#### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AELGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

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**Revision date** 23-Aug-2022

**Revision Note** Reformatted and updated existing information.

**Disclaimer**

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**End of Safety Data Sheet**